

Xuanyu Li

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EDUCATION

Cornell University

Ph. D, Department of Mathematics

Ithaca, NY, US

2023.9-Present

University of Science and Technology of China(USTC)

Undergraduate, School of Mathematical Science

Hefei, Anhui, China

2019.8-2023.7

RESEARCH INTERESTS

Minimal surfaces and harmonic maps, especially their singularity structures and existence theorem.

PREPRINTS AND PUBLICATIONS

Li X. Optimal regularity of stable harmonic maps to spheres. arXiv preprint arXiv:2608.20272 (2026).

Gaia, F. and Li, X. Existence of constant mean curvature surfaces with controlled topology in 3-manifolds. arXiv preprint arXiv:2602.16635 (2026).

Li, X. Existence of expanding harmonic map flows to hemispheres. arXiv preprint arXiv:2602.08932 (2026).

Li, X. Energy identity for Ginzburg-Landau approximation of harmonic maps. arXiv preprint arXiv:2503.23675 (2025).

Li, X. Minkowski content estimates for generic area minimizing hypersurfaces. Calc. Var. 63, 176 (2024).

AWARDS AND GRANTS

Simons Dissertation Fellowship in Mathematics

Simons Foundation, 2026

ICM 2026 Travel Grants for U.S.-Based Graduate Students

American Mathematical Society (AMS), 2026

Robert John Bättig Graduate Prize

Cornell, 2025

Guo MoRuo Scholarship (highest distinction at USTC)

USTC, 2023

Service

I have reviewed a paper for Journal of Differential Geometry.

TALKS

Generic regularity of harmonic maps

AMS 2026 Fall Western Sectional Meeting (20 mins), Arizona State University

11/2026

Uniqueness and non-uniqueness results of harmonic map flow

PDE seminar, USTC

06/2026

Union College Mathematics Conference 2026 (20 mins)

09/2026

Existence of CMC surfaces with bounded genus

Analysis Seminar, Cornell

09/2026

Geometric Analysis Seminar, CUHK

07/2026

Geometric PDES and minimal surfaces workshop, Shanghai Center for Mathematical Science

07/2026

Geometry-Topology seminar, UPenn

03/2026

Enegy identity for Ginzburg-Landau approximation of harmonic maps

Geometric Analysis seminar, USTC

06/2025

Geometric Analysis seminar, Zhejiang University

06/2025

TEACHING

Math4130 Honors Introduction to Analysis I (Grader)	<i>2026 Fall</i>
Math6160 Partial differential equations (Grader)	<i>2026 Spring</i>
Math3040 Prove it! (Recitation TA)	<i>2026 Spring</i>
Math6520 Differential manifolds (Recitation TA)	<i>2025 Fall</i>
Math4180 Complex analysis (Grader)	<i>2025 Spring</i>
Math2930 Differential equations for engineers (Recitation TA)	<i>2024 Fall</i>
Math2930 Differential equations for engineers (Recitation TA)	<i>2024 Spring</i>